Amit Chandran

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Multidisciplinary systems engineer with experience developing C++ firmware, low-latency telemetry pipelines, and fully integrated electro-mechanical systems. Proven ability to deliver high-precision software and hardware solutions, including leveraging AI developer tools safely and effectively.

Education

UCL - MEng Mechanical Engineering **Expected June 2028**

- Relevant Modules: Machine Learning, Dynamics, Control Systems
- Current Average: 78% - First Class

Experience

Systems Engineering Intern – Leonardo UK **Summer 2025 and 2026**

- Coordinated 10 interns to integrate antenna, firmware, hardware, and software into a continuous RF-to-web telemetry pipeline
- Developed a compression algorithm reducing FFT data bandwidth requirements by over **2500×**, minimising network utilisation whilst preserving critical information
- Automated manual RF signal analysis in MATLAB and reduced complexity from $\mathcal{O}(N^2)$ to $\mathcal{O}(N)$, cutting simulation time by **99.3%**
- Launched a student-led cross-functional collaboration initiative, now incorporated into the internship program for future cohorts

Projects / Extracurricular

Quadruped QDD Actuator Driver **June 2026 – Present**

- Designed a bespoke multi-layer PCB with an onboard STM32 and 3-phase inverter to drive robotic actuators, achieving a cost per unit around **70%** lower than commercial alternatives
- Developed custom C++ control algorithms for the STM32 to execute closed-loop **20 kHz** Field-Oriented Control (FOC), providing a modular framework for precise torque and position control
- Integrated the CAN protocol into the firmware, allowing for rapid telemetry exchange across 12 drivers
- Architected a multi-tiered safety system combining hardware-interrupt fault detection and SEGGER RTT logging to safely isolate overcurrent, undervoltage, and CAN faults

Technical Lead and Drone Controls Engineer – UCL Rover Team **January 2025 – Present**

- Programmed an edge-computing vision pipeline in Python and OpenCV on a Raspberry Pi to track ArUco markers in real-time, with an average pose error of $<$**1cm**
- Established a robust UDP telemetry link to stream positional data seamlessly from the edge device to a high-level MATLAB/Simulink control model
- Built a UART-based MAVLink telemetry architecture connecting Simulink models directly to an STM32H743 flight controller for real-time flight control

Smart Pet Feeder – BLE IoT System **October 2025 – March 2026**

- Developed a smart pet feeder using ESP32-S3 C++ firmware, combining a stepper motor and load cell to dispense food portions with **±1% mass accuracy** in a closed-loop system
- Implemented an event-driven BLE GATT server (Nordic UART Service) for mobile time-synchronization, eliminating external RTC costs; open-sourced firmware and DfMA CAD models
- Interfaced the feeder with an Android app via BLE, enabling remote feeding and real-time weight monitoring

Skills

- **Programming Languages** – C, C++, Python, MATLAB
- **Tools / Software** – EasyEDA (PCB), Git, Simulink, Fusion 360, Azure DevOps
- **Manufacturing / Lab** – Oscilloscopes, Soldering / SMD Rework, 3D Printing, Laser Cutting